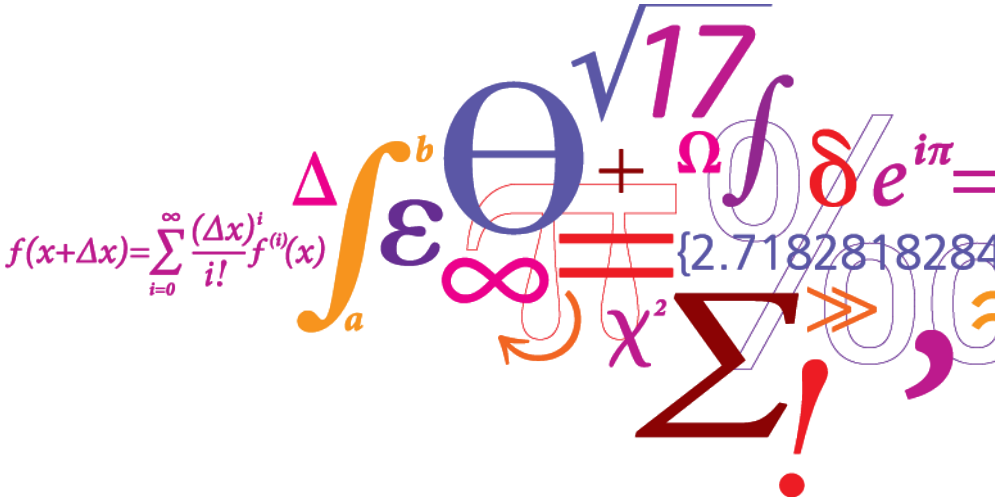
46755 – Renewables in Electricity Markets

Course introduction

Lesia Mitridati

February 2, 2026

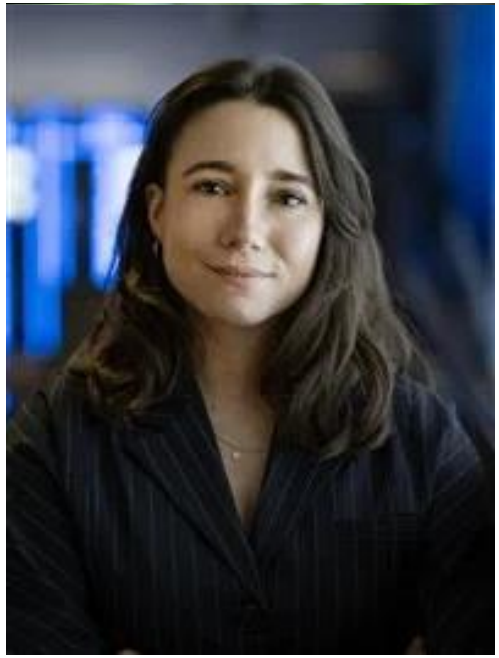


Get to know your teachers and classmates

Who are we?

Course Responsible
(Spring 2026)

Teacher
(Spring 2026)



Julia Matinez
Assistant Professor,
DTU Wind

Julia Manolopoulou
Professor, Head of Section,
DTU Wind

Who are we?

TA

Thomas Mitsopoulos
MSc Sustainable Energy Systems
s242855@student.dtu.dk

TA

Katrine Schwartz Pedersen
MSc Sustainable Energy Systems
s203701@student.dtu.dk

TA

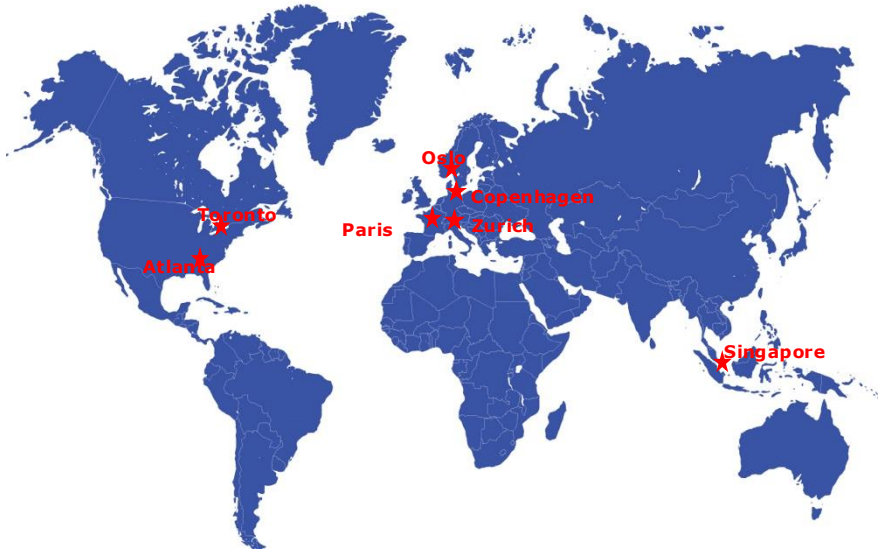
Carlos Canoyra Calero
MSc Sustainable Energy Systems
s250421@student.dtu.dk

TA

Valdemar Søgaard
PhD student at DTU Wind
vasog@dtu.dk

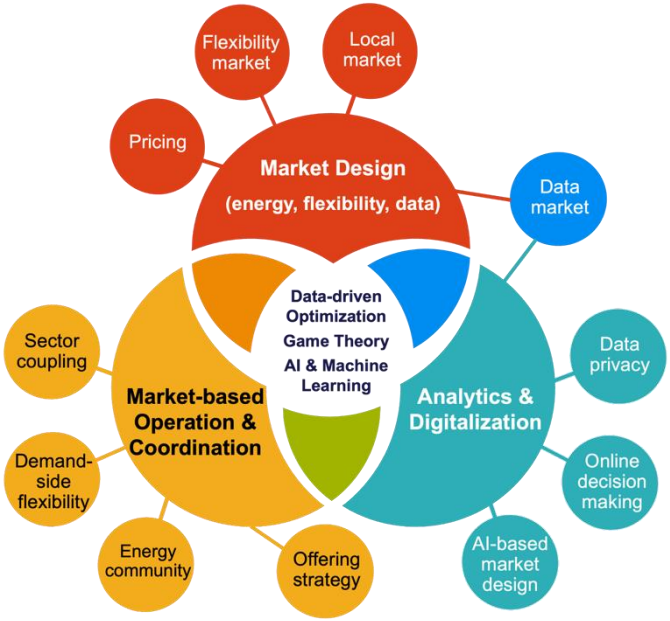
A bit about me…

- **Assistant professor, DTU Wind, EMA Section**
- Ph.D. in Electrical Engineering, DTU Elektro (2019)
- Postdocs at Georgia Tech and ETH Zurich



My research interests:

Market & information design to coordinate actions of decentralized agents towards a "social choice", using optimization, game theory and ML

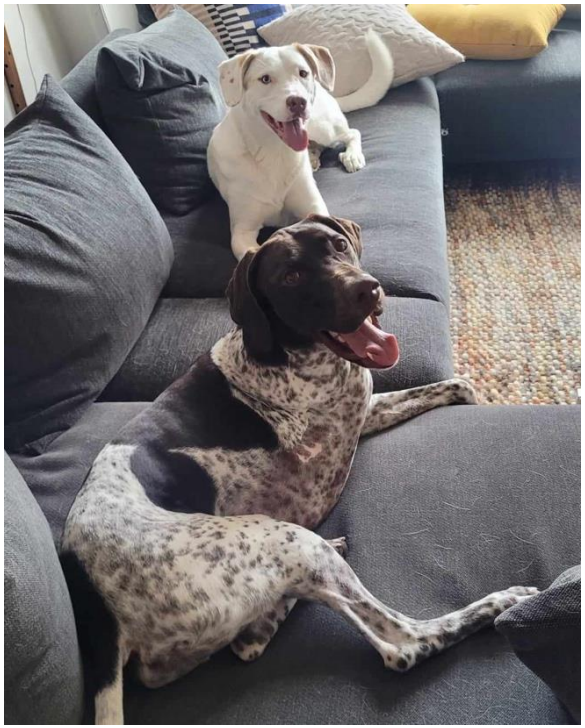


A bit about me…

I like bouldering & lead climbing…

… but I am very clumsy (so I am often injured)

I have 2 dogs (Aspen and Lichen)…

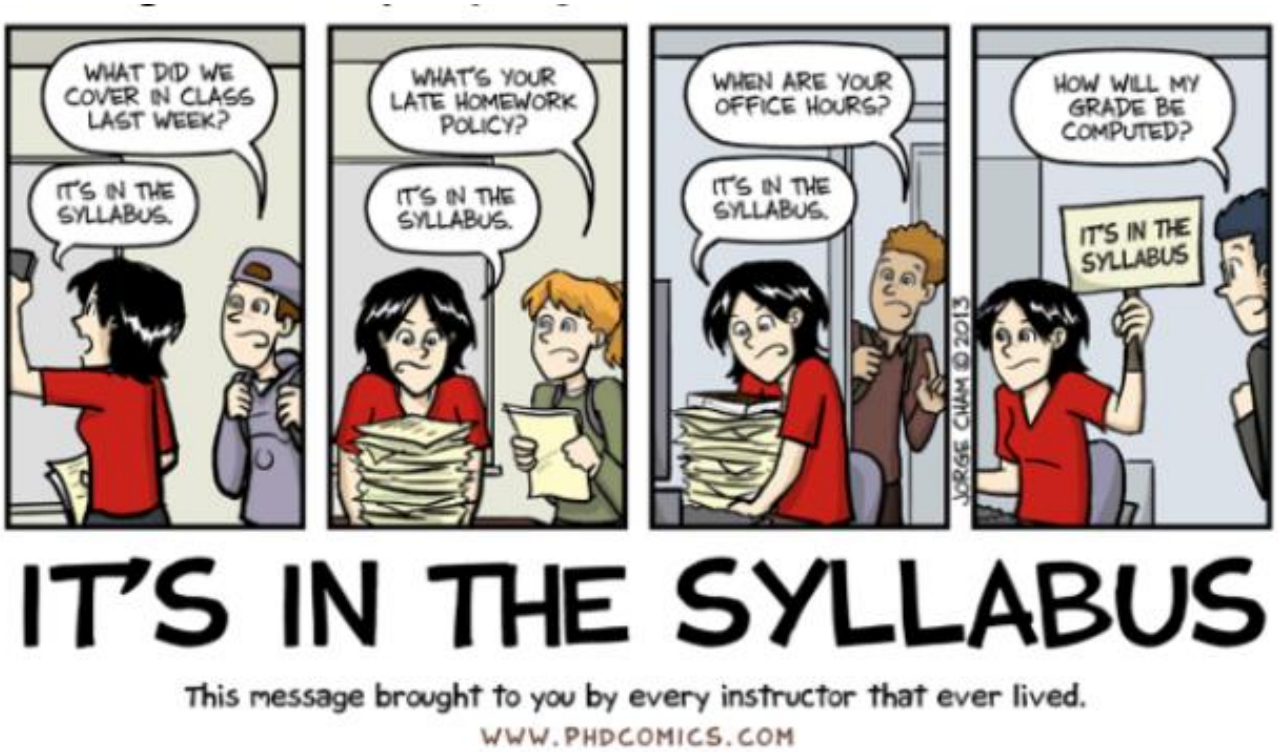


… and I like spending time with them more than with most people

I have ADHD, so I have many many many weird hobbies…

… and unfinished projects

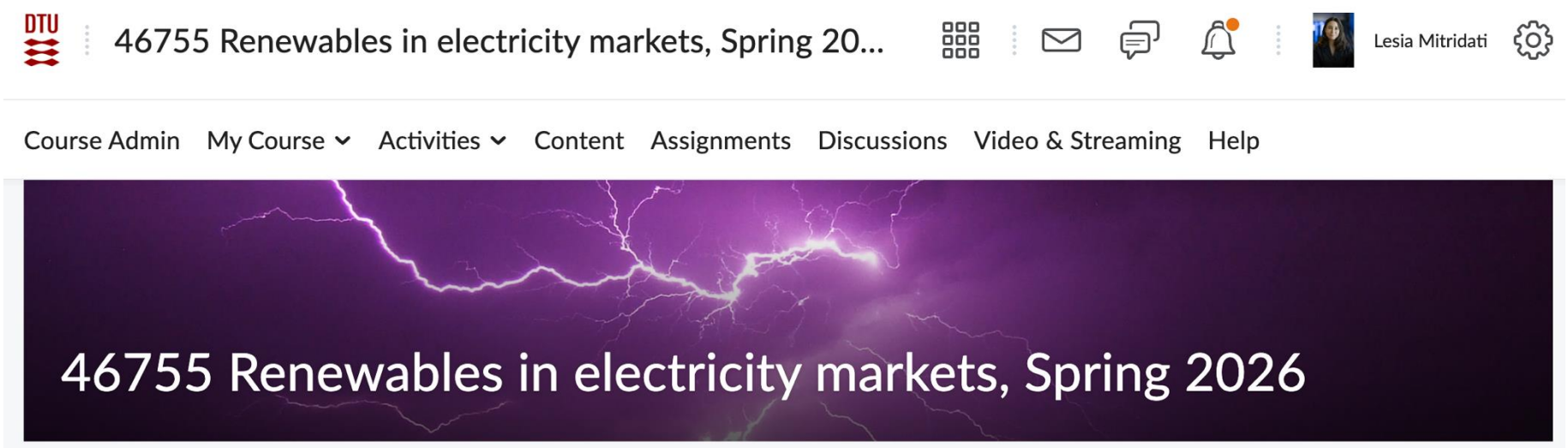
About the course content & structure



This, of course, is a joke, so please do not hesitate to reach out with any course-related questions!!

(see next slices about communication channels)

DTU Learn account is active!



Link: https://learn.inside.dtu.dk/d2l/home/294912

2 types of sessions

Asynchronous sessions - Online

- ≅**2 hours,** on your own time
- **Flipped classroom lectures:**
  - Learning objectives
  - Video lecture
  - Summary
- Other preparation instructions & optional course materials

Live sessions – In-person

- **1:00 - 2:00 PM: Lecture recap** (key concepts, examples, Kahoot, Q&A)
- **2:00 - 2:15 PM:** Break
- **2:15 - 5:00 PM: Hands-on activities** (exercises, assignments, Q&A)

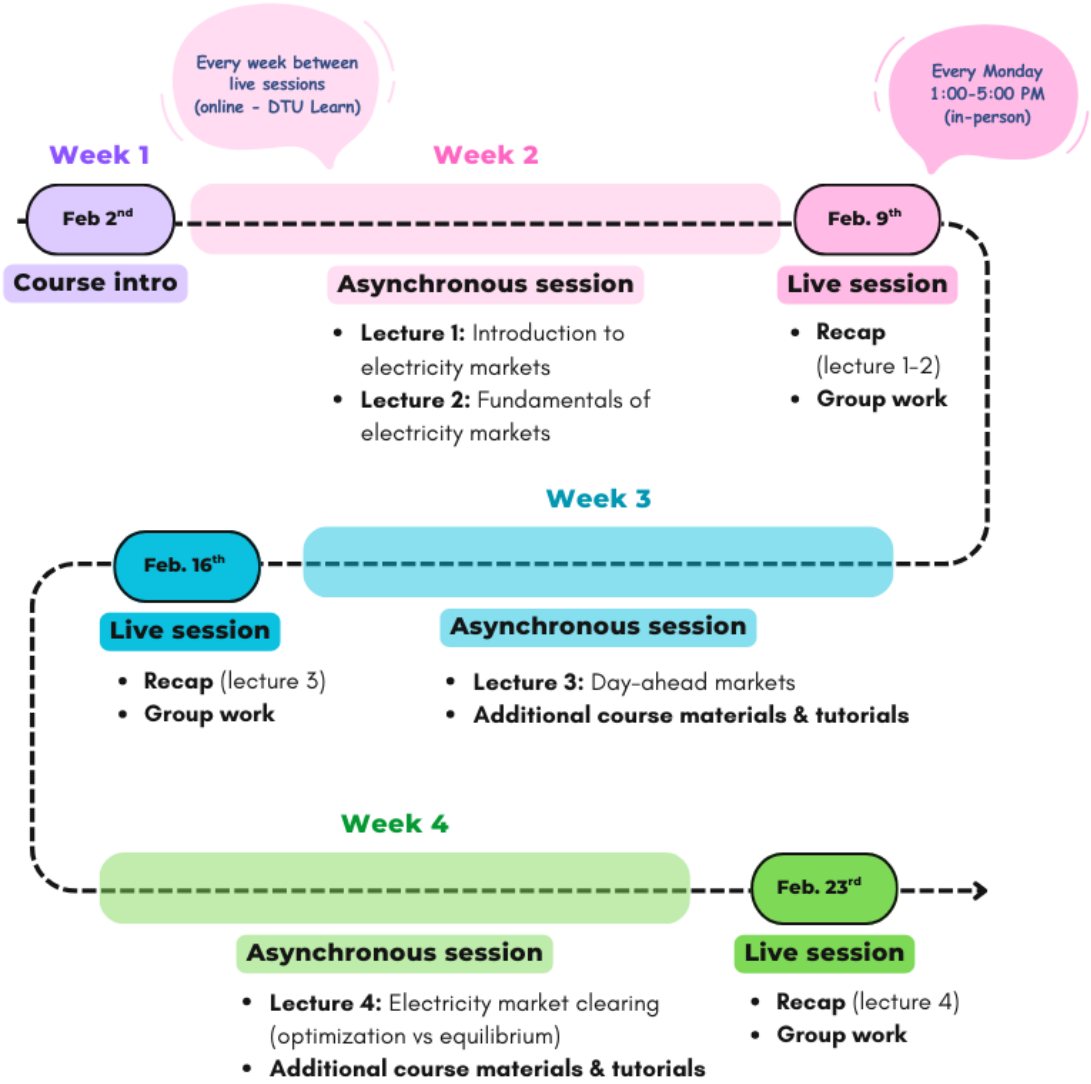
2 types of sessions

Asynchronous sessions - Online

- ≅**2 hours,** on your own time
- **Flipped classroom lectures:**
  - Learning objectives
  - Video lecture
  - Summary
- Other preparation instructions & optional course materials

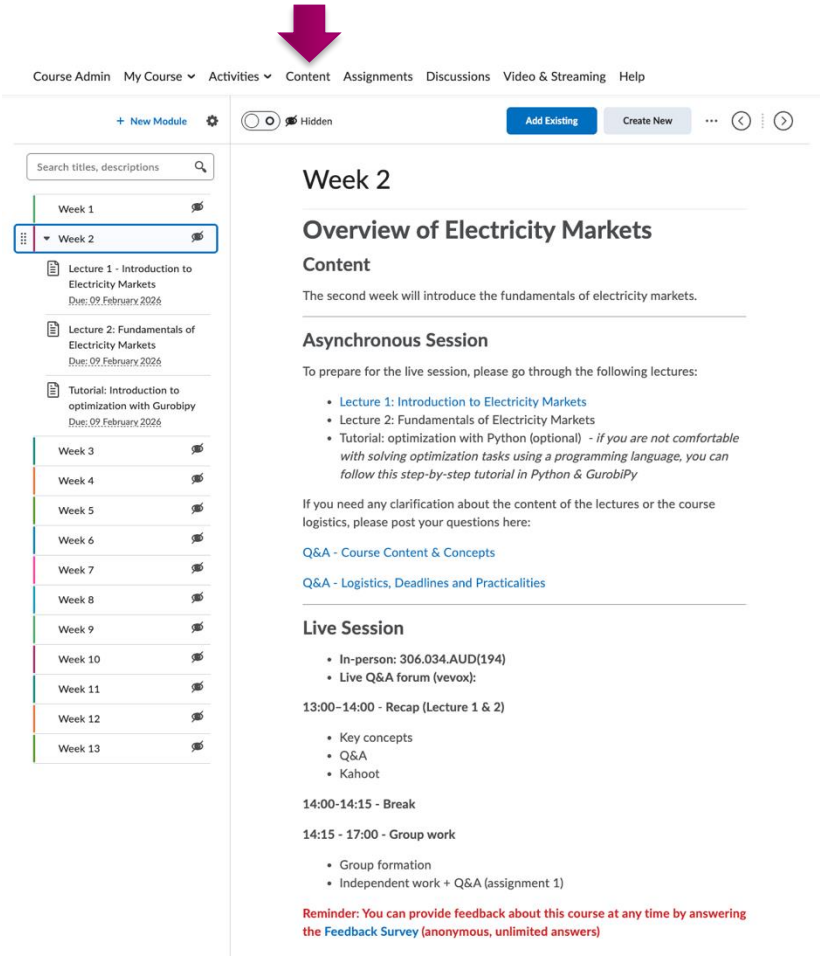
Live sessions – In-person

- **1:00 - 2:00 PM: Lecture recap** (key concepts, examples, Kahoot, Q&A)
- **2:00 - 2:15 PM:** Break
- **2:15 - 5:00 PM: Hands-on activities** (exercises, assignments, Q&A)



Where to find course materials & lecture plans?

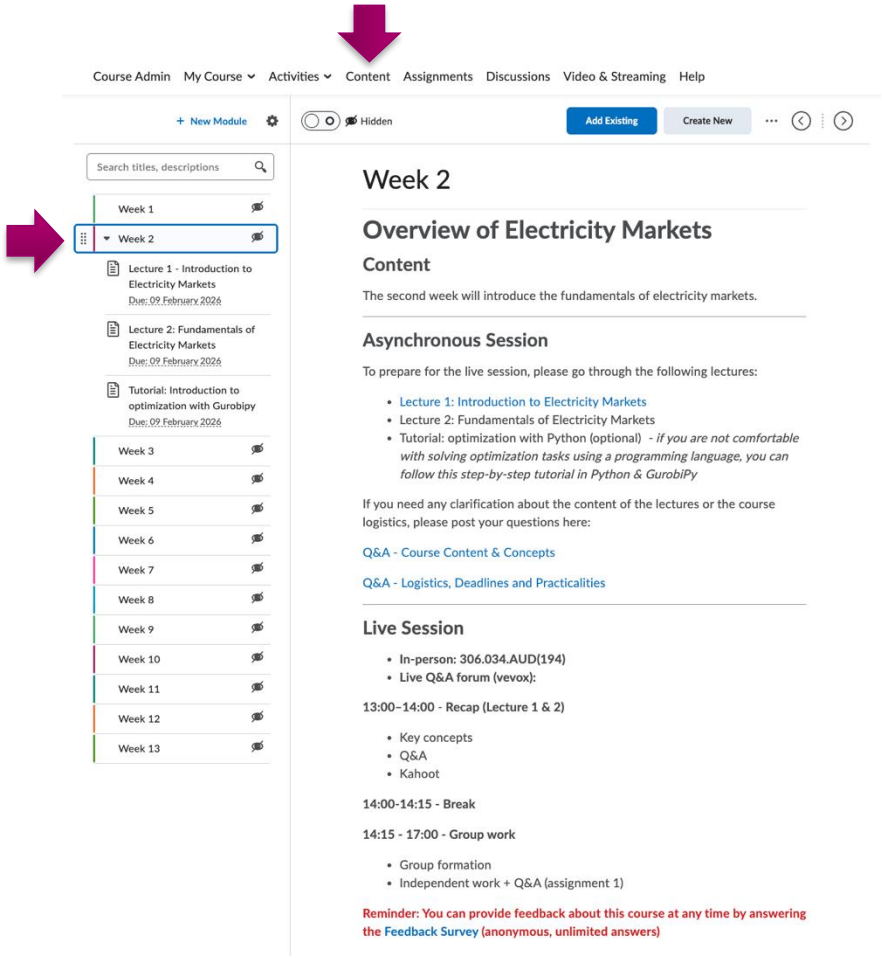
- **All course materials & lecture plans published on DTU Learn (organized per week)**



Where to find course materials & lecture plans?

- **All course materials & lecture plans published on DTU Learn (organized per week)**

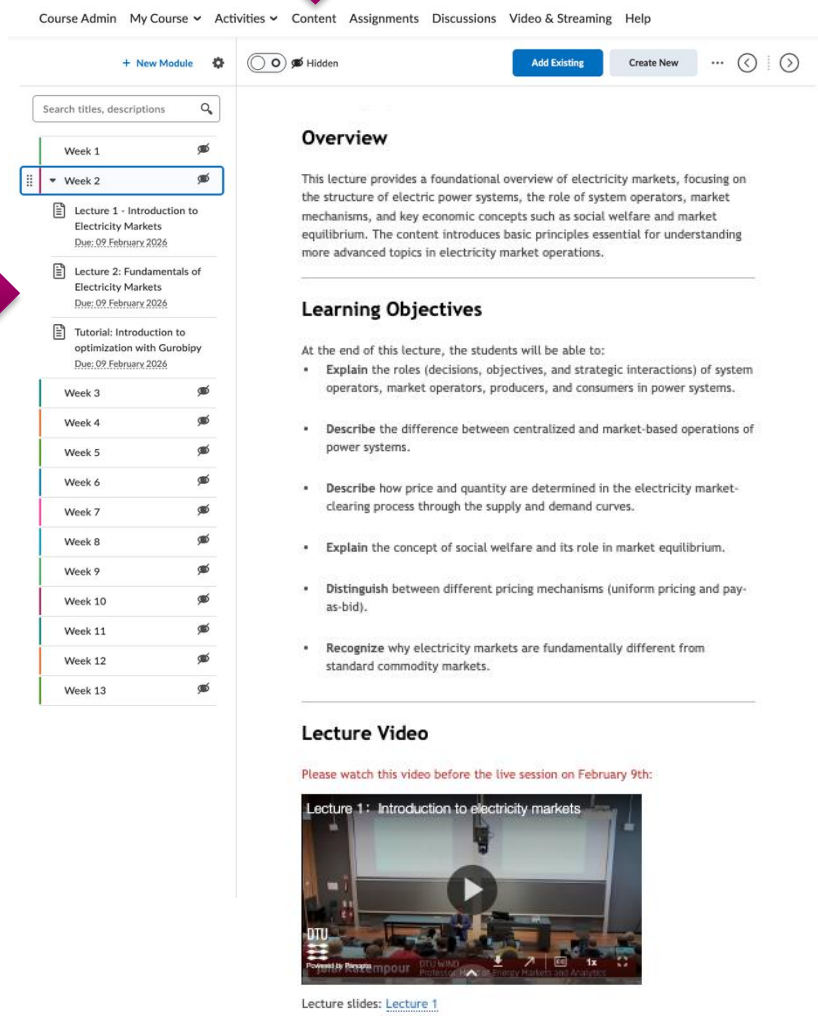
  - Lecture plan for the week published 1 week in advance (asynchronous + live session) – *Week 2 available now*



Where to find course materials & lecture plans?

- **All course materials & lecture plans published on DTU Learn (organized per week)**

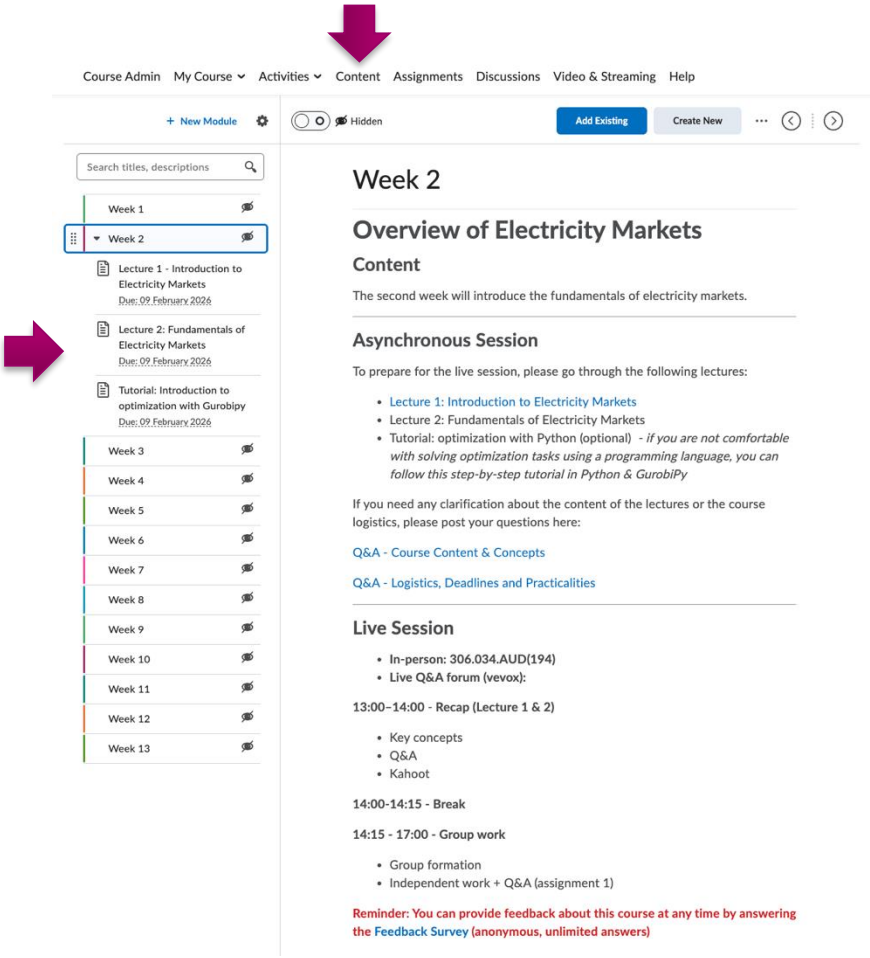
  - Lecture plan for the week published 1 week in advance (asynchronous + live session) – *Week 2 available now*
  - Course materials for asynchronous sessions published 1 week in advance - *Week 2 available now*



Where to find course materials & lecture plans?

- **All course materials & lecture plans published on DTU Learn (organized per week)**

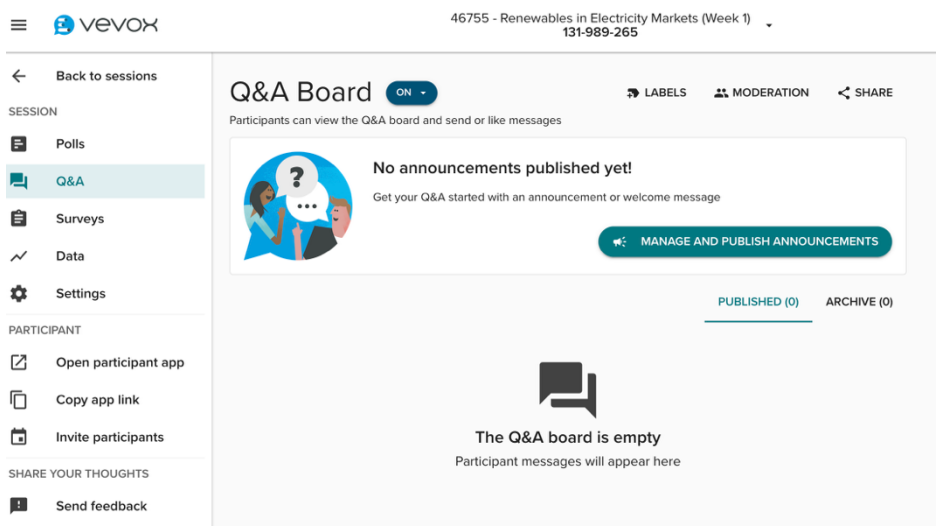
  - Lecture plan for the week published 1 week in advance (asynchronous + live session) – *Week 2 available now*
  - Course materials for asynchronous sessions published 1 week in advance - *Week 2 available now*
  - Course materials for live session published on the same day - *Week 2 available on February 9th at 13:00*



When/where to ask questions?

During live sessions

- 1:00 – 2:00 PM: questions related to course content & logistics (in-class or through live Vevox Q&A)
- 2:15 – 5 PM: questions related to assignments and specific to each group's progress



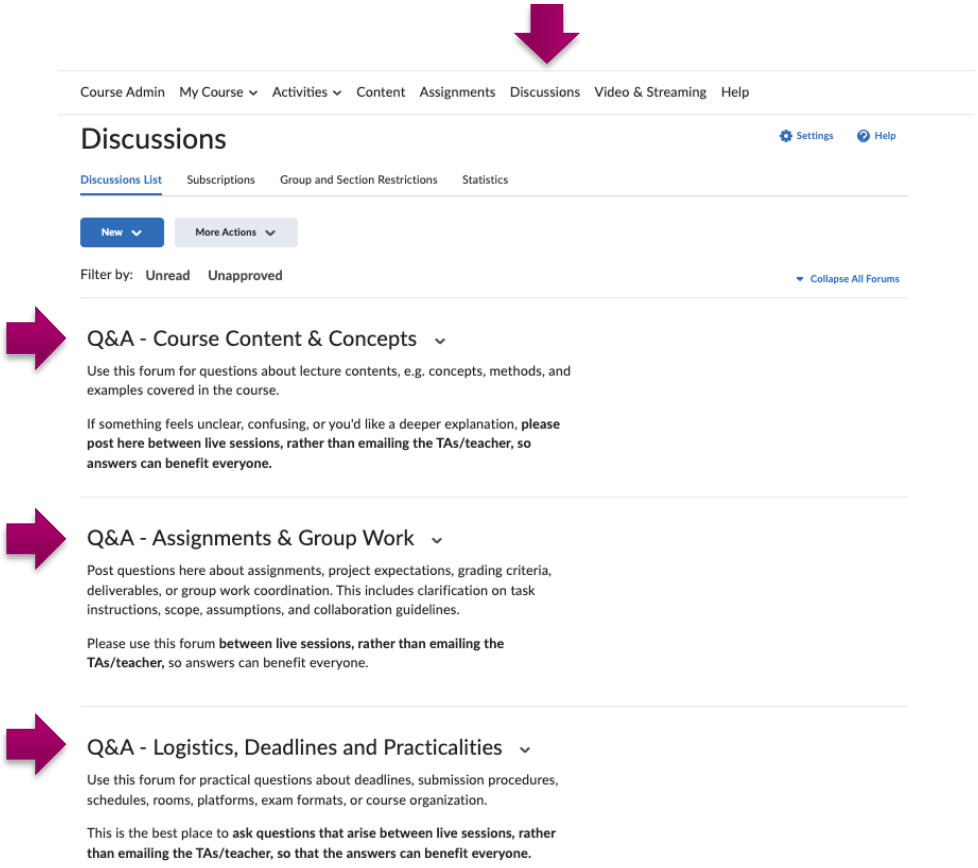
When/where to ask questions?

During live sessions

- 1:00 – 2:00 PM: questions related to course content & logistics (in-class or through live Vevox Q&A)
- 2:15 – 5 PM: questions related to assignments and specific to each group's progress

Between live sessions

- DTU Learn Q&A forums: questions related to course content, logistics or assignments



When/where to ask questions?

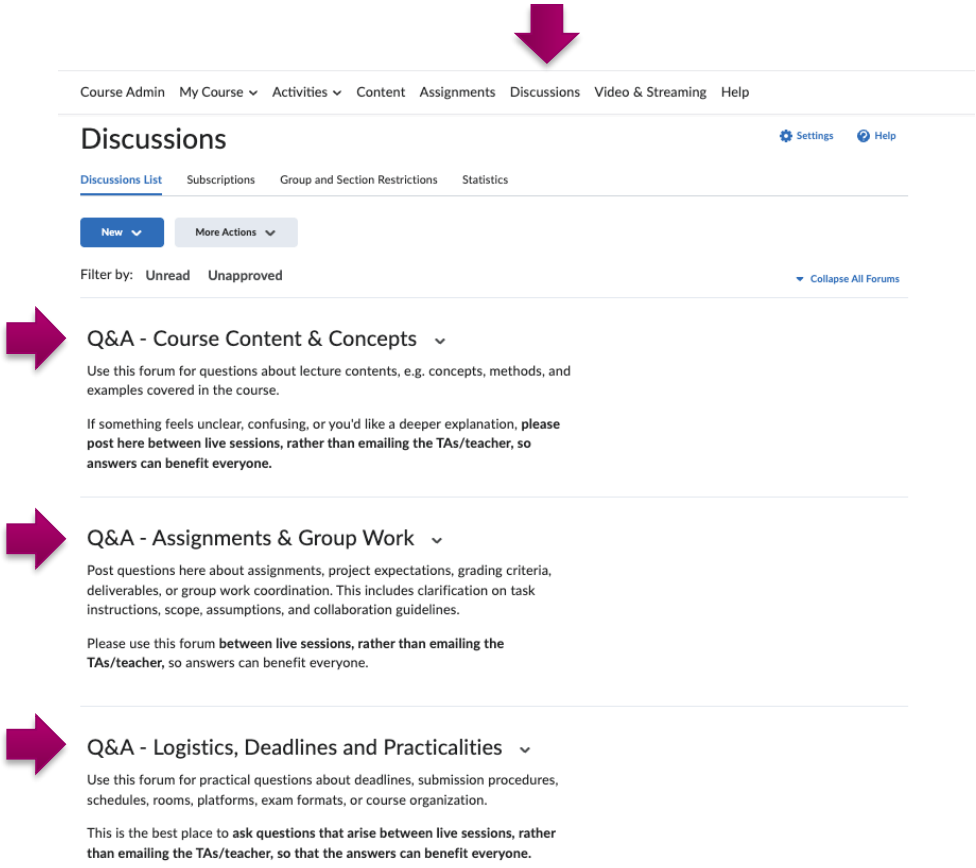
During live sessions

- 1:00 – 2:00 PM: questions related to course content & logistics (in-class or through live Vevox Q&A)
- 2:15 – 5 PM: questions related to assignments and specific to each group's progress

Between live sessions

- DTU Learn Q&A forums: questions related to course content, logistics or assignments

Also consider these forums as a **communication platform among students**, helping each other and brainstorming ideas, and try to actively participate in discussions



When/where to ask questions?

During live sessions

- 1:00 – 2:00 PM: questions related to course content & logistics (in-class or through live Vevox Q&A)
- 2:15 – 5 PM: questions related to assignments and specific to each group's progress

Between live sessions

- DTU Learn Q&A forums: questions related to course content, logistics or assignments

When you have a question:

- **Ask it as soon as possible** (instead of waiting for the break on Mondays or an entire week for the following live session)
- **Avoid sending a direct email to the teacher/TAs**
- Instead, ask it "publicly", i.e. in-class, through the live vevox Q&A or the DTU Learn forums, so that **all students can benefit from the answer**
- **Reserve the breaks for coffee & the hands-on sessions for questions specific to your own group's assignment progress**

General course objectives

- ✓ Electricity markets are today central in the way we exchange electricity.
- ✓ The increasing penetration of renewable energy sources with stochastic and variable supply in power systems has a substantial and complex impact on electricity markets.

Link: https://kurser.dtu.dk/course/46755

General course objectives

- ✓ Electricity markets are today central in the way we exchange electricity.
- ✓ The increasing penetration of renewable energy sources with stochastic and variable supply in power systems has a substantial and complex impact on electricity markets.
- ✓ The objective of this course is to introduce the students to the fundamental aspects of renewable energy in electricity markets, from the point of view of both market operators and market participants.

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

1) Explain the fundamentals of electricity markets

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

1) Explain the fundamentals of electricity markets
2) Explain various markets, actors, and pricing mechanisms

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

1) Explain the fundamentals of electricity markets
2) Explain various markets, actors, and pricing mechanisms
3) Explain the market-clearing process from both optimization and equilibrium perspectives

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

1) Explain the fundamentals of electricity markets
2) Explain various markets, actors, and pricing mechanisms
3) Explain the market-clearing process from both optimization and equilibrium perspectives
4) Describe the impact of renewables on electricity markets

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

1) Explain the fundamentals of electricity markets
2) Explain various markets, actors, and pricing mechanisms
3) Explain the market-clearing process from both optimization and equilibrium perspectives
4) Describe the impact of renewables on electricity markets
5) Develop mathematical models including optimization and equilibrium problems for clearing various electricity markets

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

1) Explain the fundamentals of electricity markets
2) Explain various markets, actors, and pricing mechanisms
3) Explain the market-clearing process from both optimization and equilibrium perspectives
4) Describe the impact of renewables on electricity markets
5) Develop mathematical models including optimization and equilibrium problems for clearing various electricity markets
6) Design mathematical models for market participants enabling them to make optimal offering strategy decisions under uncertainty

Link: https://kurser.dtu.dk/course/46755

Learning objectives

Eight learning objectives:

A student who has met the objectives of the course will be able to:

1) Explain the fundamentals of electricity markets
2) Explain various markets, actors, and pricing mechanisms
3) Explain the market-clearing process from both optimization and equilibrium perspectives
4) Describe the impact of renewables on electricity markets
5) Develop mathematical models including optimization and equilibrium problems for clearing various electricity markets
6) Design mathematical models for market participants enabling them to make optimal offering strategy decisions under uncertainty
7) Implement some of these mathematical models in Julia or Python

Link: https://kurser.dtu.dk/course/46755

Learning objectives

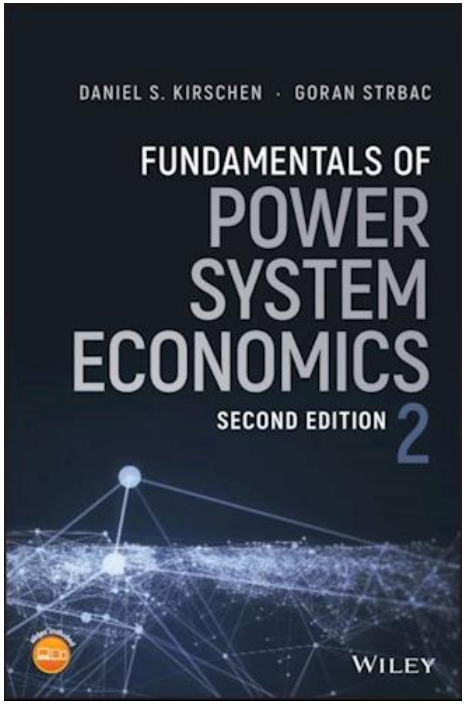
Eight learning objectives:

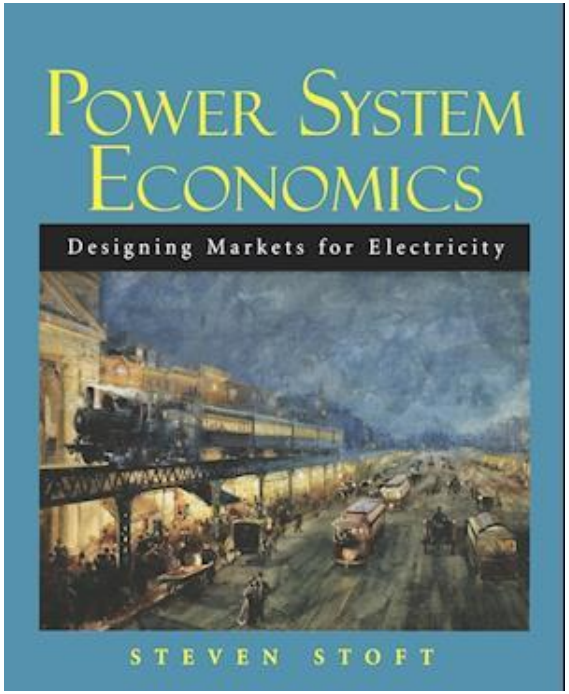
A student who has met the objectives of the course will be able to:

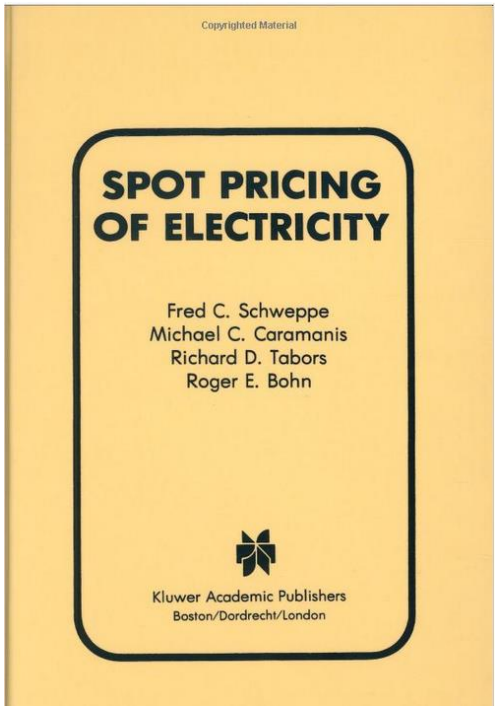
1) Explain the fundamentals of electricity markets
2) Explain various markets, actors, and pricing mechanisms
3) Explain the market-clearing process from both optimization and equilibrium perspectives
4) Describe the impact of renewables on electricity markets
5) Develop mathematical models including optimization and equilibrium problems for clearing various electricity markets
6) Design mathematical models for market participants enabling them to make optimal offering strategy decisions under uncertainty
7) Implement some of these mathematical models in Julia or Python
8) Present, analyze, and interpret the numerical results

Link: https://kurser.dtu.dk/course/46755

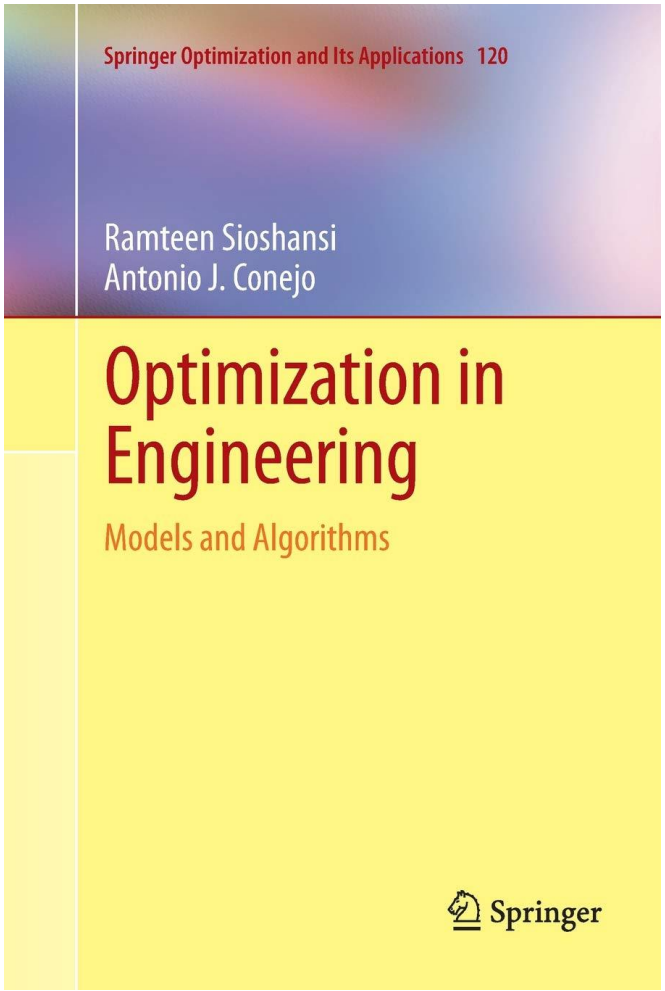
Recommended books on electricity markets



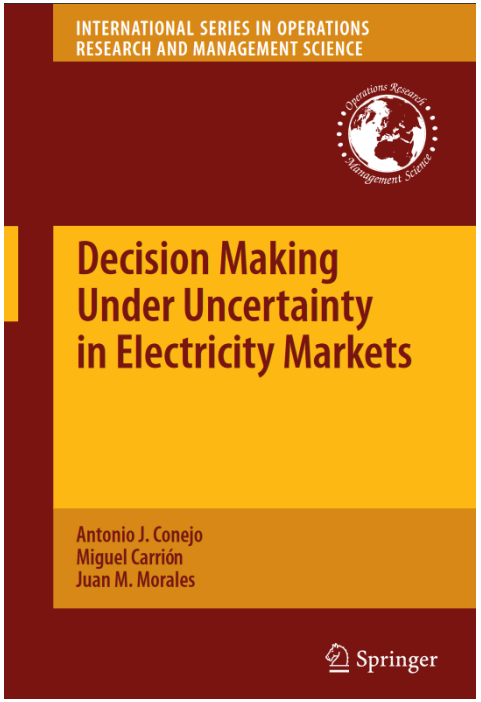


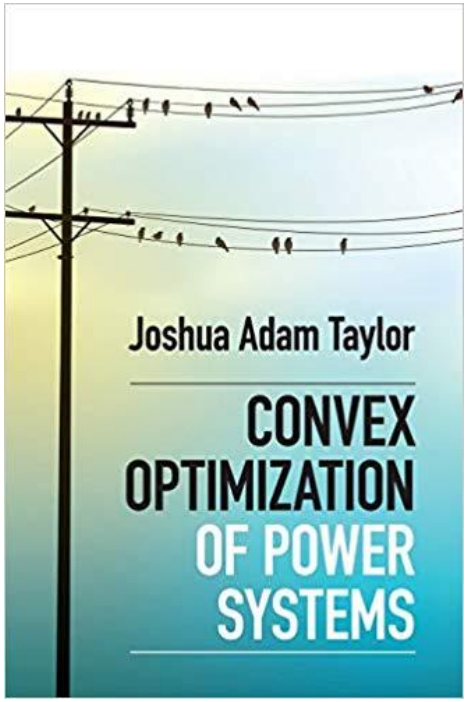


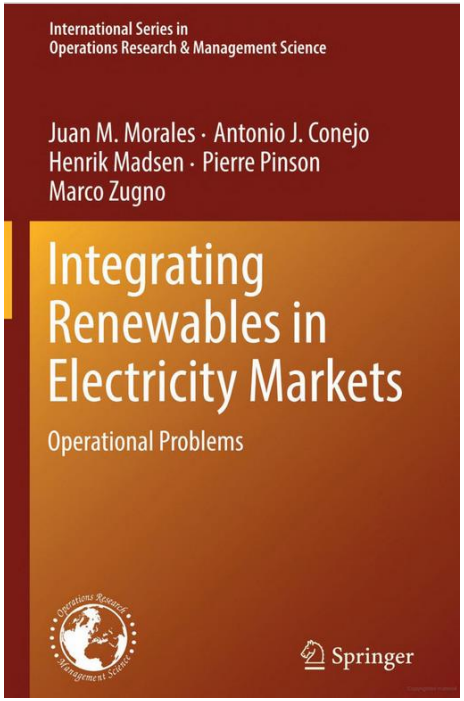
For the ”optimization” part of the course

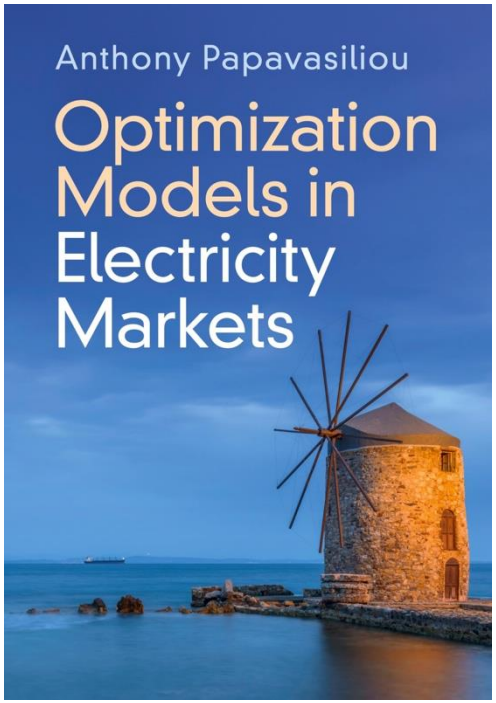


More top books in electricity markets, power system optimization, and beyond









and many others …

What are you most excited to learn about in this course?

What are you most excited to learn about in this course?

RESULTS SLIDE

Course schedule - Spring 2026

Week # (date)	Asynchronous Session (preparation)	Live Session	Deadlines & Updates on the assignments
#1 (Feb 2)	**NA**	Course introduction	
#2 (Feb 9)	• **Lecture 1:** Introduction to electricity markets • **Lecture 2:** Fundamentals of electricity markets	• Recap (Lecture 1 & 2) • Group work (assignment 1)	**Assignment 1** to be published
#3 (Feb 16)	• **Lecture 3:** Day-ahead markets	• Recap (Lecture 3) • Group work (assignment 1)	**Deadline** for group formation
#4 (Feb 23)	**Lecture 4:** Electricity market clearing: Optimization vs. equilibrium	• Recap (Lecture 4) • Group work (assignment 1)	
#5 (Mar 2)	**Lecture 5:** Intraday and balancing markets	• Recap (Lecture 5) • Group work (assignment 1)	
#6 (Mar 9)	**Lecture 6:** Ancillary service markets	• Recap (Lecture 6) • Group work (assignment 1)	
#7 (Mar 16)	**Lecture 7:** Impact of renewables on electricity markets	• Recap (Lecture 7) • Group work (assignment 1)	
#8 (Mar 23)	**No Lecture**	• Group work (assignment 1)	**Deadline** for Assignment 1
Easter			
#9 (April 6)	**Lecture 8:** Offering strategy of wind power producers	• Recap (Lecture 8) • Group work (assignment 2)	**Assignment 2** to be published
#10 (Apr 13)	**Lecture 9:** Risk-constrained offering strategy	• Recap (Lecture 9) • Group work (assignment 2)	
#11 (Apr 20)	**Lecture 10:** Offering strategy in ancillary service markets	• Recap (Lecture 10) • Group work (assignment 2)	
#12 (April 27)	**Bonus lecture: topic of your choice**	• **Bonus lecture: topic of your choice** • Group work (assignment 2)	
#13 (May 4)	**No Lecture**	• Group work (assignment 2)	**Deadline** for Assignment 2

Course schedule - Spring 2026

Week # (date)	Asynchronous Session (preparation)	Live Session	Deadlines & Updates on the assignments
#1 (Feb 2)	**NA**	Course introduction	
#2 (Feb 9)	• **Lecture 1:** Introduction to electricity markets •	• Recap (Lecture 1 [illegible]) [illegible]ssignment 1)	**Assignment 1** to be published
#3 (Feb 16)	•	[illegible] 3) [illegible]ssignment 1)	**Deadline** for group formation
#4 (Feb 23)	**L** [illegible]	[illegible] 4) [illegible]ssignment 1)	
#5 (Mar 2)	**L**	[illegible] 5) [illegible]ssignment 1)	
#6 (Mar 9)	**L**	[illegible] 6) [illegible]ssignment 1)	
#7 (Mar 16)	**L** markets	[illegible] 7) • Group work (assignment 1)	
#8 (Mar 23)	**No Lecture**	• Group work (assignment 1)	**Deadline** for Assignment 1
Easter			
#9 (April 6)	**Lecture 8:** Offering strategy of wind power producers	• Recap (Lecture 8) • Group work (assignment 2)	**Assignment 2** to be published
#10 (Apr 13)	**Lecture 9:** Risk-constrained offering strategy	• Recap (Lecture 9) • Group work (assignment 2)	
#11 (Apr 20)	**Lecture 10:** Offering strategy in ancillary service markets	• Recap (Lecture 10) • Group work (assignment 2)	
#12 (April 27)	**Bonus lecture: topic of your choice**	• **Bonus lecture: topic of your choice** • Group work (assignment 2)	
#13 (May 4)	**No Lecture**	• Group work (assignment 2)	**Deadline** for Assignment 2

- **Deadlines visible in DTU Learn** (assignment & calendar)
- **Hard deadlines!** Late submission will impact your grade
- You can **submit multiple times** until deadline (only last submission kept) so submit early (and resubmit) to avoid late submission

Course schedule - Spring 2026

Week # (date)	Asynchronous Session (preparation)	Live Session	Deadlines & Updates on the assignments
#1 (Feb 2)	**NA**	Course introduction	
#2 (Feb 9)	• **Lecture 1:** Introduction to electricity markets • **Lecture 2:** Fundamentals of electricity markets	• Recap (Lecture 1 & 2) • Group work (assignment 1)	**Assignment 1** to be published
#3 (Feb 16)	• **Lecture 3:** Day-ahead markets	• Recap (Lecture 3) • Group work (assignment 1)	Deadline for group formation
#4 (Feb 23)	**Lecture 4:** Electricity market clearing: Optimization vs. equilibrium	• Recap (Lecture 4) • Group work (assignment 1)	
#5 (Mar 2)	**Lecture 5:** Intraday and balancing markets	• Recap (Lecture 5) • Group work (assignment 1)	
#6 (Mar 9)	**Lecture 6:** Ancillary service markets	• Recap (Lecture 6) • Group work (assignment 1)	
#7 (Mar 16)	**Lecture 7:** Impact of renewables on electricity markets	• Recap (Lecture 7) • Group work (assignment 1)	
#8 (Mar 23)	**No Lecture**	• Group work (assignment 1)	Deadline for **Assignment 1**
Easter			
#9 (April 6)	**Lecture 8:** Offering strategy of wind power producers	• Recap (Lecture 8) • Group work (assignment 2)	**Assignment 2** to be published
#10 (Apr 13)	**Lecture 9:** Risk-constrained offering strategy	• Recap (Lecture 9) • Group work (assignment 2)	
#11 (Apr 20)	**Lecture 10:** Offering strategy in ancillary service markets	• Recap (Lecture 10) • Group work (assignment 2)	
#12 (April 27)	**Bonus lecture: topic of your choice**	• **Bonus lecture: topic of your choice** • Group work (assignment 2)	
#13 (May 4)	**No Lecture**	• Group work (assignment 2)	Deadline for **Assignment 2**

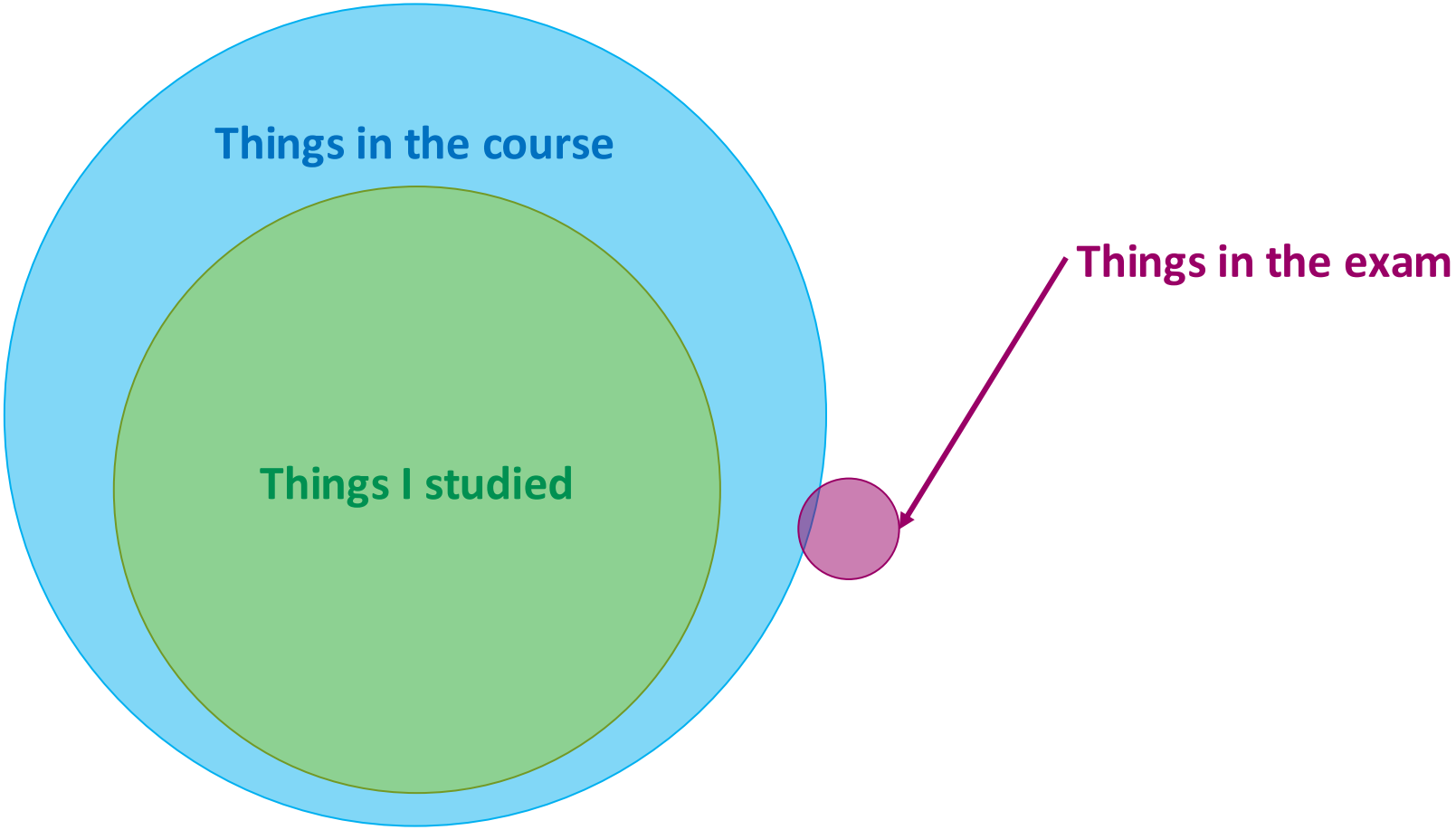
Student Agency
The power to engage in meaningful and intentional actions
Aknowledging rights and responsabilities
Voice, choice, and ownership
Developing critical thinking
Actively participating in construction our learning

What topic would you like to learn about in the "bonus" lecture?

You can pick a topic one of these topics (examples), or propose your own: capacity markets & RES support schemes, investment in RES & PPAs, PtX and hybrid power plants, privacy and cybersecurity, energy communities & P2P markets, data visualization & storytelling challenge, value-oriented forecasting, learning rival participants' bids, industry talk, no lecture & focus on assignments

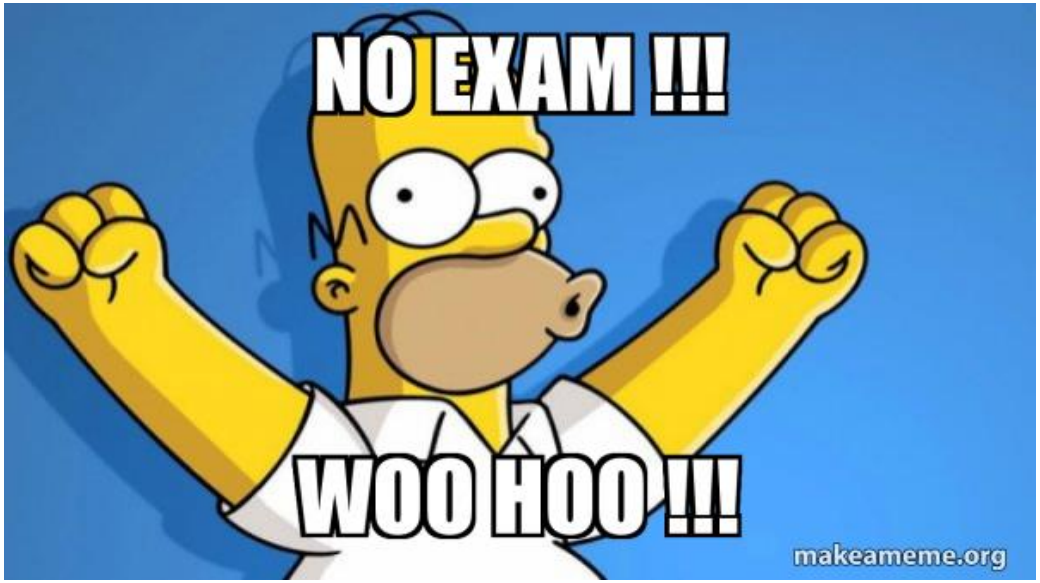
How to pass this course (and get a 12)

The issue with (most) final exams is…



How to pass this course (and get a 12)

No Final Written Exam!



Two group assignments

Two group assignments

Assignment 1

(focus on the market operator perspective)

- To be published on February 9 at 1:00 PM
- Deadline: March 23, 11:59 PM

Two group assignments

Assignment 1

(focus on the market operator perspective)

- To be published on February 9 at 1:00 PM
- Deadline: March 23, 11:59 PM

Assignment 2

(focus on the market participant perspective)

- To be published on March 23 at 1:00 PM
- Deadline: May 04, 11:59 PM

Two group assignments

Assignment 1

(focus on the market operator perspective)

- To be published on February 9 at 1:00 PM
- Deadline: March 23, 11:59 PM

Assignment 2

(focus on the market participant perspective)

- To be published on March 23 at 1:00 PM
- Deadline: May 04, 11:59 PM

✓ Both assignments to be completed in groups of **four**

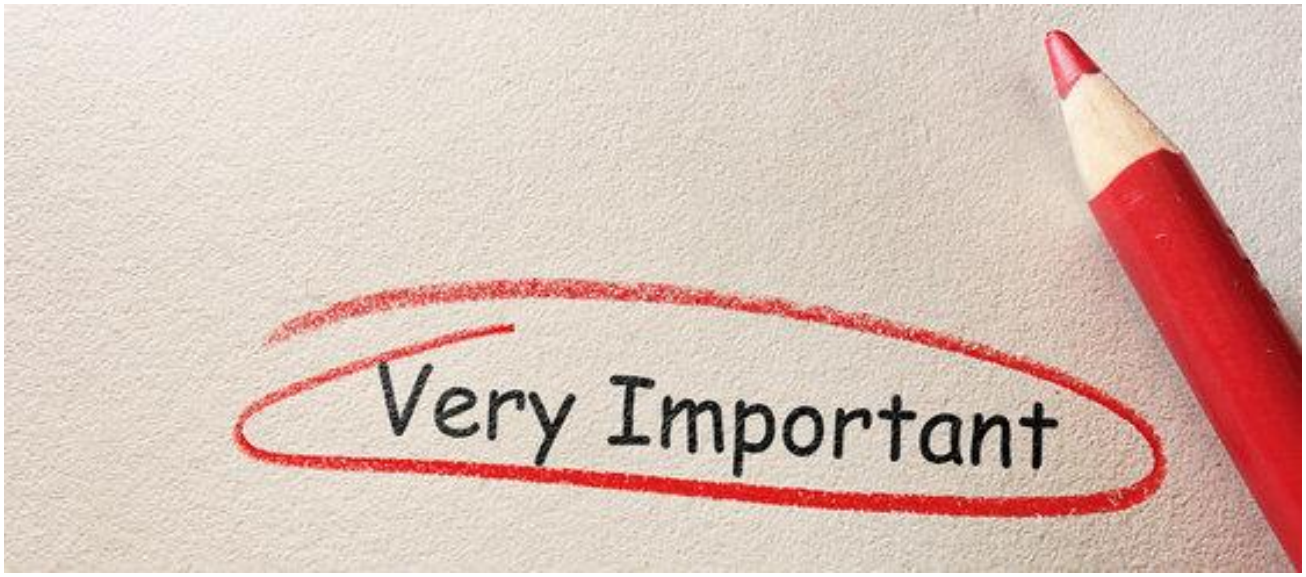
- less than 4 is **not** fine due to the large number of students in the course!

Both projects are quite time and effort demanding!

Please make sure you have good planning for projects!

Both projects are heavily programming demanding!

Require solving optimization tasks using any programming language of your choice (Julia, Python, etc)



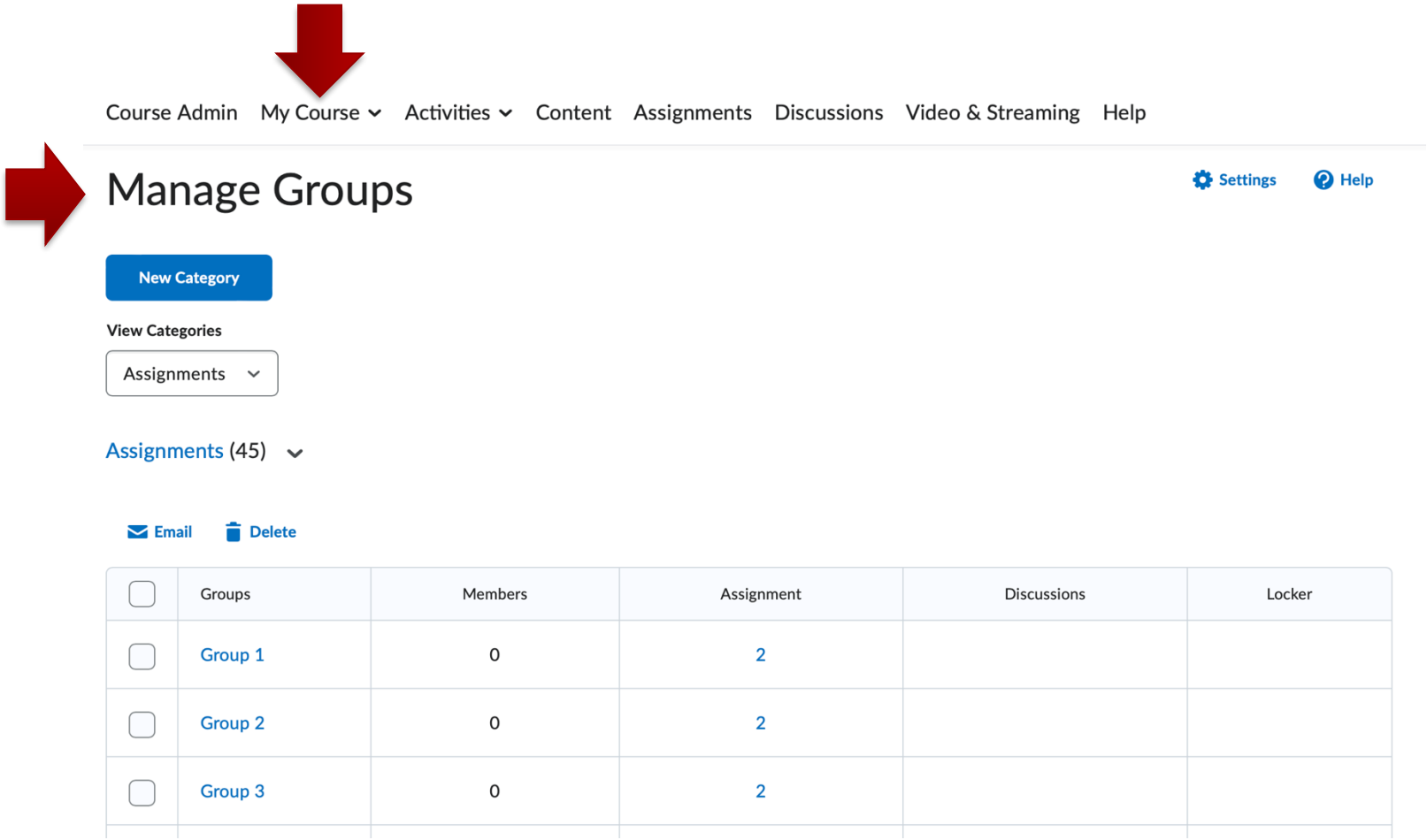
This course is <u>NOT</u> meant to support your programming skills

- If you are not already comfortable with a programming language, we strongly recommend that you <u>first</u> improve your programming skills, and take this course next year!

Check the other recommended prerequisites: https://kurser.dtu.dk/course/46755

Where to form groups?

Please form your groups on DTU Learn, ideally, by the beginning of the 2nd session (February 9, 1:00 PM), but no later than by the end of the 3rd session (February 16, 5:00 PM)



Suggestion (1/3): How to form groups?

Before finalizing your group, make sure you have a good balance of various skills!

Relevant skills may include:

- ❑ Project management and communication
- ❑ Mathematical model development (e.g., good optimization skills)
- ❑ Programming (coding)
- ❑ Analysis of numerical results and thorough discussions
- ❑ Plotting, formatting and writting

Suggestion (2/3): How to form groups?

Align your expectations in terms of grade and workload:

- Recall a 5-ECTS course requires a workload of around 140 hours.

Examples:

- If one student aims for a grade of 12 while another is ok with a 7 or 4, they may not be ideal group partners since their expectations are not well aligned.
- Aside from the 4 hours of class, if one student is willing to dedicate 8-10 hours per week to the project while the other is considering a half workload, they may not be well-matched to work together in the same group.

Suggestion (3/3): How to form groups?

Use the “group formation” forum on DTU Learn:

Q&A - Assignments & Group Work

Post questions here about assignments, project expectations, grading criteria, deliverables, or group work coordination. This includes clarification on task instructions, scope, assumptions, and collaboration guidelines.

Please use this forum **between live sessions, rather than emailing the TAs/teacher,** so answers can benefit everyone.

Topic	Threads	Posts	Last Post
Group Discussions Group/section restrictions. Private group discussion area	0	0	
Group formation **Are you looking for a group or a group member?** Please, feel free to post here! You can share your skills and what you're looking for in a group, or invite others with a specific expertise to join your group. Relevant skills may include **mathematical modelling, programming, writing, project management, data analytics, energy economics, etc.** **Note that you are responsible for forming and managing your own groups (including resolving future disagreements), so please take the time to ensure a good fit.**	0	0	

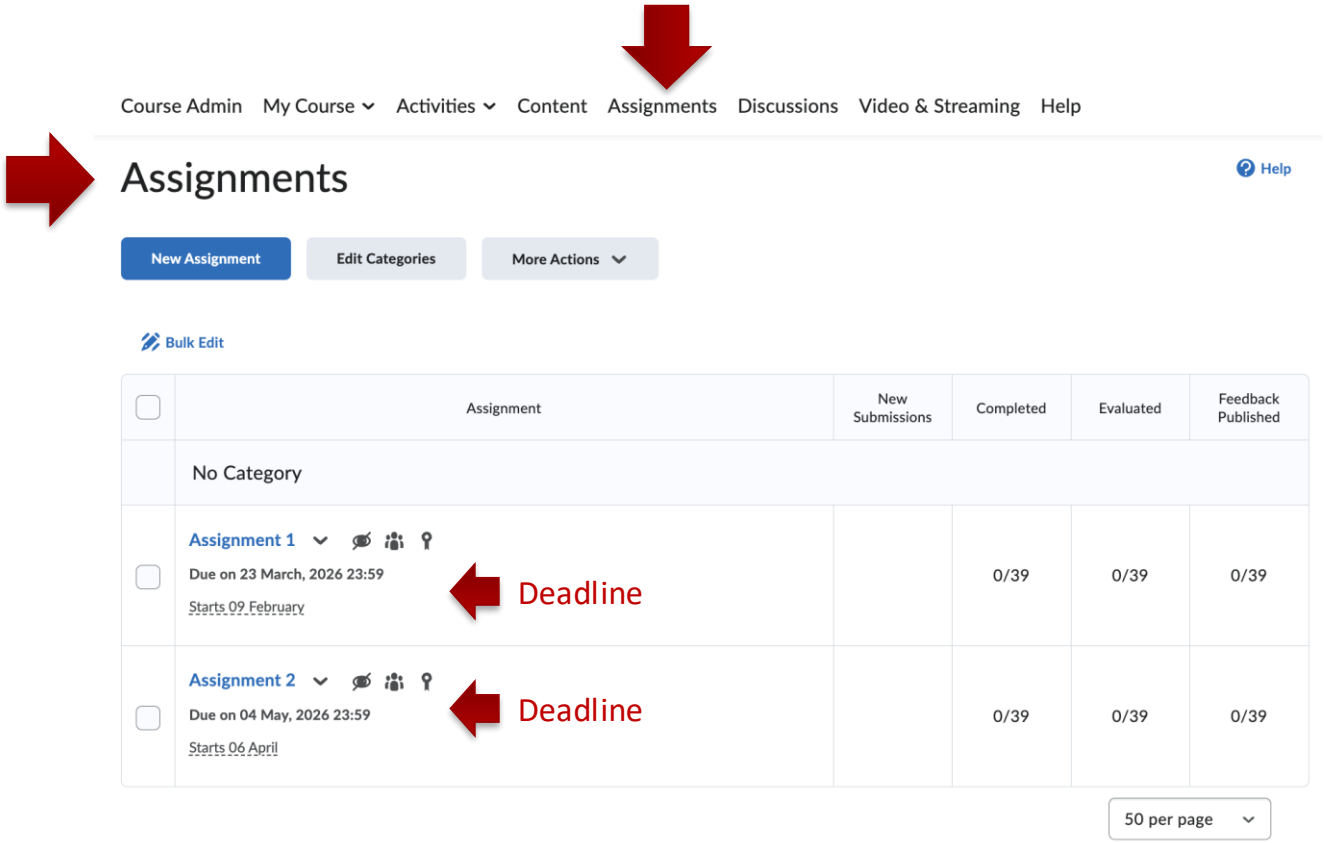
Disagreement in the group?

You are responsible for forming your own groups and resolving any potential disagreements within the group. Please take the time to ensure you create a well-functioning group.

- The teacher and TAs are NOT responsible for resolving any internal conflicts within the group.
- ONLY ONE report per group assignment should be submitted.

Where to submit group assignment reports?

On DTU Learn (one report per group per assignment):



What will we do every Monday <u>2-5pm</u>?

- **February 2nd :**
  - No group work planned
  - You can start forming your groups or preparing Lectures 1 & 2 for week 2

- **February 9th and onwards**:
  - Every group works on the assignment
  - TAs (and the lecturer) will be available for the Q&A

Assessment

Requirements:

- Submit **one report per group per assignment** (see template and content in next slides)

Assessment

Requirements:

- Submit **one report per group per assignment** (see template and content in next slides)
- Include a **participation table** in the first page of each report

	Name (group member 1)	Name (group member 2)	Name (group member 3)	Name (group member 4)
Contribution to model development	? %	? %	? %	? %
Contribution to programming	? %	? %	? %	? %
Contribution to the analysis of numerical results	? %	? %	? %	? %
Contribution to writing the report	? %	? %	? %	? %
Overall	**? %**	**? %**	**? %**	**? %**

Assessment

Requirements:

- Submit **one report per group per assignment** (see template and content in next slides)
- Include a **participation table** in the first page of each report

	Name (group member 1)	Name (group member 2)	Name (group member 3)	Name (group member 4)
Contribution to model development	? %	? %	? %	? %
Contribution to programming	? %	? %	? %	? %
Contribution to the analysis of numerical results	? %	? %	? %	? %
Contribution to writing the report	? %	? %	? %	? %
Overall	**? %**	**? %**	**? %**	**? %**

Ideally, each group member should have similar contribution to every task. We highly encourage students to have at least 15% contribution to every task, **but this is not part of grading**

Assessment

Requirements:

- Submit **one report per group per assignment** (see template and content in next slides)
- Include a **participation table** in the first page of each report

	Name (group member 1)	Name (group member 2)	Name (group member 3)	Name (group member 4)
Contribution to model development	? %	? %	? %	? %
Contribution to programming	? %	? %	? %	? %
Contribution to the analysis of numerical results	? %	? %	? %	? %
Contribution to writing the report	? %	? %	? %	? %
Overall	**? %**	**? %**	**? %**	**? %**

Overall individual participation in each assignment influences the grade: Any value lower than 25% will lower the individual grade proportionally (see next slides).

Assessment

Requirements:

- Submit **one report per group per assignment** (see template and content in next slides)
- Include a **participation table** in the first page of each report

	Name (group member 1)	Name (group member 2)	Name (group member 3)	Name (group member 4)
Contribution to model development	? %	? %	? %	? %
Contribution to programming	? %	? %	? %	? %
Contribution to the analysis of numerical results	? %	? %	? %	? %
Contribution to writing the report	? %	? %	? %	? %
Overall	**? %**	**? %**	**? %**	**? %**

- Provide a **well-documented code** (commented, read-me file with all dependencies, data & other files needed to run the code and reproduce results)

Report template and content

- Single column, A4, 11pt, pdf format

Report template and content

- Single column, A4, 11pt, pdf format
- Recommended length **12 pages** - maximum **15 pages** (each report). Efficient writing is part of the assessment criteria, so the report should be as short and efficient as possible

Report template and content

- Single column, A4, 11pt, pdf format
- Recommended length **12 pages** - maximum **15 pages** (each report). Efficient writing is part of the assessment criteria, so the report should be as short and efficient as possible
- A short introduction can be included.

Report template and content

- Single column, A4, 11pt, pdf format
- Recommended length **12 pages** - maximum **15 pages** (each report). Efficient writing is part of the assessment criteria, so the report should be as short and efficient as possible
- A short introduction can be included.
- Mathematical formulations should be complete, with all notations defined rigorously. If relevant, a compact representation can be used.

Report template and content

- Single column, A4, 11pt, pdf format
- Recommended length **12 pages** - maximum **15 pages** (each report). Efficient writing is part of the assessment criteria, so the report should be as short and efficient as possible
- A short introduction can be included.
- Mathematical formulations should be complete, with all notations defined rigorously. If relevant, a compact representation can be used.
- It is recommended to provide only the most important numerical results in the form of figures or tables. Please avoid reporting trivial results. It is highly important to analyze the results obtained and derive relevant & concise insights.

Report template and content

- Single column, A4, 11pt, pdf format
- Recommended length **12 pages** - maximum **15 pages** (each report). Efficient writing is part of the assessment criteria, so the report should be as short and efficient as possible
- A short introduction can be included.
- Mathematical formulations should be complete, with all notations defined rigorously. If relevant, a compact representation can be used.
- It is recommended to provide only the most important numerical results in the form of figures or tables. Please avoid reporting trivial results. It is highly important to analyze the results obtained and derive relevant & concise insights.
- You can include additional relevant information as needed, such as case study setup, # of variables and constraints, computational aspects (machinery used, time, etc).

Report template and content

- Single column, A4, 11pt, pdf format
- Recommended length **12 pages** - maximum **15 pages** (each report). Efficient writing is part of the assessment criteria, so the report should be as short and efficient as possible
- A short introduction can be included.
- Mathematical formulations should be complete, with all notations defined rigorously. If relevant, a compact representation can be used.
- It is recommended to provide only the most important numerical results in the form of figures or tables. Please avoid reporting trivial results. It is highly important to analyze the results obtained and derive relevant & concise insights.
- You can include additional relevant information as needed, such as case study setup, # of variables and constraints, computational aspects (machinery used, time, etc).
- An appendix can be included (outside the page limit) for less important information, e.g., input data, etc.

Report template and content

- Single column, A4, 11pt, pdf format
- Recommended length **12 pages** - maximum **15 pages** (each report). Efficient writing is part of the assessment criteria, so the report should be as short and efficient as possible
- A short introduction can be included.
- Mathematical formulations should be complete, with all notations defined rigorously. If relevant, a compact representation can be used.
- It is recommended to provide only the most important numerical results in the form of figures or tables. Please avoid reporting trivial results. It is highly important to analyze the results obtained and derive relevant & concise insights.
- You can include additional relevant information as needed, such as case study setup, # of variables and constraints, computational aspects (machinery used, time, etc).
- An appendix can be included (outside the page limit) for less important information, e.g., input data, etc.

A sample report will be available on DTU Learn on February 9th (together with Assignment 1).

Assessment criteria

Group-based evaluation of each assignment, based on following criteria (same for both assignments):

- Mathematical model development: 35%
- Code(s): 25%
- Report: efficient writing, rigorous results, and rigorous discussions: 40%

A detailed rubric table will be shared along with each assignment, so you will have an idea how your report will be graded.

Grading

Based on <u>group assignments (report and code):</u>

- Assignment 1: 0 - 100 points (50% of final grade)
- Assignment 2: 0 - 100 points (50% of final grade)

Grading

Based on <u>group assignments (report and code):</u>

- Assignment 1: 0 - 100 points (50% of final grade) x participation coeff - assignment 1
- Assignment 2: 0 - 100 points (50% of final grade) x participation coeff - assignment 2

Individual participation coefficient

- Between 0 and 1 for each assignment
- will not be more than 1 if the contribution is more than 25%.

Grading

Based on <u>group assignments (report and code)</u>:

- Assignment 1: 0 - 100 points (50% of final grade) x participation coeff - assignment 1
- Assignment 2: 0 - 100 points (50% of final grade) x participation coeff - assignment 2

Individual participation coefficient

- Between 0 and 1 for each assignment
- will not be more than 1 if the contribution is more than 25%.

Conversion of final <u>individual grade</u> (0-100%) to the 7-scale Danish system (based on DTU instructions):

- [92% - 100%] ---> Grade 12 (pass)
- [84% - 92%) ---> Grade 10 (pass)
- [68% - 84%) ---> Grade 7 (pass)
- [60% - 68%) ---> Grade 4 (pass)
- [50% - 60%) ---> Grade 2 (pass)
- [20% - 50%) ---> Grade 0 (fail)
- [0% - 20%) ---> Grade -3 (fail)

Grading - Example

Consider a group with the following report grades:

Assignment 1: 88%

Assignment 2: 82%

Grading - Example

Consider a group with the following report grades:

Assignment 1: 88%

Assignment 2: 82%

Student A has an overall contribution of 25% to Assignment 1, and 25% to Assignment 2.

Grading - Example

Consider a group with the following report grades:

Assignment 1: 88%

Assignment 2: 82%

Student A has an overall contribution of 25% to Assignment 1, and 25% to Assignment 2.

- Grade: { [0.88 * 1.0] + [0.82 * 1.0] } /2 = 85%

Individual participation coefficients

Grading - Example

Consider a group with the following report grades:

Assignment 1: 88%

Assignment 2: 82%

Student A has an overall contribution of 25% to Assignment 1, and 25% to Assignment 2.

- Grade: { [0.88 * 1.0] + [0.82 * 1.0] } /2 = 85%

Individual participation coefficients

- [92% - 100%] ---> Grade 12 (pass)
- [84% - 92%) ---> Grade 10 (pass)
- [68% - 84%) ---> Grade 7 (pass)
- [60% - 68%) ---> Grade 4 (pass)
- [50% - 60%) ---> Grade 2 (pass)
- [20% - 50%) ---> Grade 0 (fail)
- [0% - 20%) ---> Grade -3 (fail)

Grading - Example

Consider a group with the following report grades:

Assignment 1: 88%

Assignment 2: 82%

Student B has an overall contribution of 20% to Assignment 1, and 35% to Assignment 2.

Grading - Example

Consider a group with the following report grades:

Assignment 1: 88%

Assignment 2: 82%

Student B has an overall contribution of 20% to Assignment 1, and 35% to Assignment 2.

- Grade: { [0.88 * 0.8] + [0.82 * 1.0] } /2 = 76.2%

Individual participation coefficients

Grading - Example

Consider a group with the following report grades:

Assignment 1: 88%

Assignment 2: 82%

Student B has an overall contribution of 20% to Assignment 1, and 35% to Assignment 2.

- Grade: { [0.88 * 0.8] + [0.82 * 1.0] } /2 = 76.2%

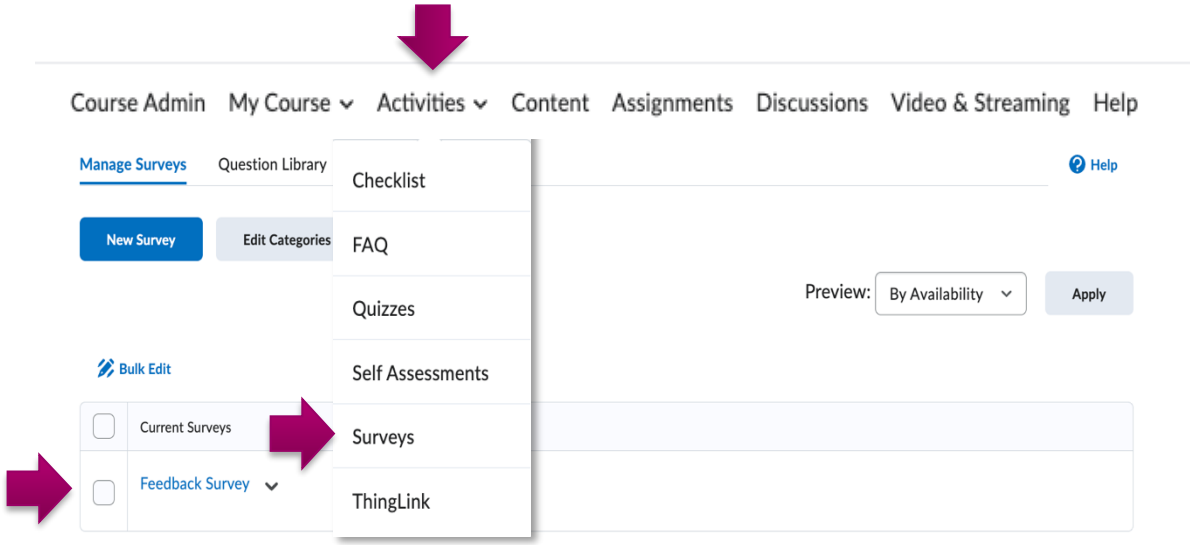
Individual participation coefficients

- [92% - 100%] ---> Grade 12 (pass)
- [84% - 92%) ---> Grade 10 (pass)
- [68% - 84%) ---> Grade 7 (pass)
- [60% - 68%) ---> Grade 4 (pass)
- [50% - 60%) ---> Grade 2 (pass)
- [20% - 50%) ---> Grade 0 (fail)
- [0% - 20%) ---> Grade -3 (fail)

Final thoughts & feedback

Please help us improve this course!

- We always welcome & are grateful to receive feedback:
  - **In-person, or using the live Vevox polls or Q&A forum during class**
  - **Feedback Survey (DTU Learn):** anonymous, unlimited answers



- **Concrete examples are helpful!**
- We will do our best to address all feedback as soon as possible (during the semester when possible)

Feedback

Take a moment to reflect on the following

Strongly disagree **Strongly agree**

- ✓ *I know what to learn and where to get the information needed*
- ✓ *I feel safe to express myself, ask questions, get and give feedback*
- ✓ *I feel confident in my ability to achieve the learning objectives of this course*
- ✓ *I feel confident using the tools needed (python, DTU learn, etc)*

Feedback

Please share any additional feedback:

e.g.: Is anything still unclear? What should we keep doing? What can we do better next time? Anything else that comes to mind...

Thanks for your attention & your participation!

How to contact me?

- **For any general course-related question:** on DTU Learn - Q&A forums (most convenient & fastest)
- **For individual/private questions:**
  - ✓ By email lemitri@dtu.dk – mention "46755" in email subject (may take a few days to reply)
  - ✓ On Teams (faster)